Arista Access Node ELEC-ACC-01 QoS Congestion Event Resolution SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4967

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve Quality of Service (QoS) congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on Arista Access Node ELEC-ACC-01, located at the Electronic City Hub. The equipment in question runs on the Arista CLI dialect. The root cause of these issues lies in the interaction between the QoS policy configuration and the underlying hardware, specifically the scheduling and queuing mechanisms employed by the Arista 7500 Series. The Arista 7500 Series uses a multi-level queuing architecture, where packets are assigned to one of eight priority queues based on the DSCP (DiffServ Code Point) value in the IP header. Packets are then dequeued and sent out on the corresponding egress interface based on the priority of the queue. However, if the QoS policy configuration is not optimized, packets may be placed in high-priority queues, leading to congestion and microbursts. Additionally, if the interface is experiencing high utilization, packets may be tail-dropped, further exacerbating the congestion. The QoS queue starvation occurs when packets from a particular traffic class are consistently placed in a low-priority queue, leading to long delays and packet loss. To resolve these issues, it is essential to optimize the QoS policy configuration, adjust the scheduling and queuing mechanisms, and ensure that the interface utilization is within acceptable limits.

Pre-requisite Checks

- Check: Verify that the redundant link is active and functioning correctly.

- Check: Ensure that the remote hands are dispatched to the Electronic City Hub for any necessary physical interventions.

- Check: Perform a thorough review of the QoS policy configuration to identify any potential issues.

Potential Root Causes

Cause	Probability
Inadequate QoS policy configuration	High
Suboptimal scheduling and queuing mechanisms	Medium
High interface utilization	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy, Expected Output: Displays the current QoS policy configuration, including the priority queues and scheduling algorithms used., Description: This command will help identify any potential issues with the QoS policy configuration, such as misconfigured priority queues or incorrect scheduling algorithms.
```

```
Command: show interface statistics, Expected Output: Displays the
interface statistics, including the number of packets transmitted and
received, as well as the number of packets dropped., Description: This
command will help identify any issues with the interface, such as high
utilization or packet drops, which may be contributing to the
congestion event.

Command: show queue statistics, Expected Output: Displays the queue
statistics, including the number of packets in each queue and the
average queue length., Description: This command will help identify
any issues with the queuing mechanisms, such as queue starvation or
high queue lengths, which may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Optimize the QoS policy configuration, Command: configure qos policy , Description: This step involves reviewing and optimizing the QoS policy configuration to ensure that packets are correctly assigned to priority queues and that the scheduling algorithms are optimal. This may involve adjusting the DSCP values, priority queue assignments, and scheduling algorithms to ensure that packets are transmitted efficiently and without congestion.

Step 2: Step: Adjust the scheduling and queuing mechanisms, Command: configure queue , Description: This step involves adjusting the scheduling and queuing mechanisms to ensure that packets are transmitted efficiently and without congestion. This may involve adjusting the priority queue assignments, burst sizes, and scheduling algorithms to ensure that packets are transmitted without packet drops or queue starvation.

Step 3: Step: Reduce interface utilization, Command: configure interface , Description: This step involves reducing the interface utilization to prevent congestion and packet drops. This may involve adjusting the interface bandwidth, configuring Quality of Service (QoS) policies, or implementing traffic shaping to ensure that packets are transmitted efficiently and without congestion.

Post-Check Verification

```
Command: show qos policy, Expected Output: Displays the updated QoS
policy configuration, including the priority queues and scheduling
algorithms used., Description: This command will help verify that the
QoS policy configuration has been optimized and that packets are
correctly assigned to priority queues.

Command: show interface statistics, Expected Output: Displays the
updated interface statistics, including the number of packets
transmitted and received, as well as the number of packets dropped.,
Description: This command will help verify that the interface
utilization has been reduced and that packets are transmitted
efficiently without congestion or packet drops.

Command: show queue statistics, Expected Output: Displays the updated
queue statistics, including the number of packets in each queue and
the average queue length., Description: This command will help verify
that the queuing mechanisms have been optimized and that packets are
transmitted efficiently without congestion or queue starvation.
```

Roll-back Procedures

- Step: Restore the original QoS policy configuration, Command: configure qos policy , Description: This step involves restoring the original QoS policy configuration to its previous state. This will involve reverting the changes made to the QoS policy configuration, including the priority queue assignments and scheduling algorithms.

- Step: Restore the original interface configuration, Command: configure interface , Description: This step involves restoring the original interface configuration to its previous state. This will involve

reverting the changes made to the interface bandwidth and QoS policies.

- Step: Restore the original queuing mechanisms, Command: configure queue , Description: This step involves restoring the original queuing mechanisms to their previous state. This will involve reverting the changes made to the priority queue assignments, burst sizes, and scheduling algorithms.